

Internship - Project 3 (Final Project) Assignments

Machine Learning & Data Science | Unsupervised Learning

K-Means

Hierarchical

DBSCAN

PCA

Isolation Forest

Student Name	Project Title	Algorithms
Amanullah Fazil S	Customer Value Segmentation & Retention Analysis	K-Means + Hierarchical Clustering
Udayanidhi S	Wholesale Buyer Profile Discovery	K-Means + Principal Component Analysis
Muthu Sankar V	Dry Bean Morphology Pattern Discovery	Hierarchical Clustering + DBSCAN
Lakshana R	Bank Customer Campaign Segmentation	K-Means + DBSCAN
M Mugesh	Human Activity Pattern Discovery from Smartphone Sensors	Principal Component Analysis + K-Means
Nitesh Varshan K K	Network Intrusion Anomaly Discovery	Isolation Forest + DBSCAN
Palani Bharathi	Wine Chemical Profile Segmentation & Outlier Detection	K-Means + Isolation Forest
Nithish T	Tumor Feature Structure Discovery	Principal Component Analysis + Hierarchical Clustering
S Ramkumar	Household Energy Usage Pattern & Anomaly Detection	Hierarchical Clustering + Isolation Forest

Each student is assigned exactly two primary unsupervised-learning algorithms. Labels, when available, must not be used during training and may be used only for post-hoc validation. Submit code, dashboard, outputs, report, and a short demonstration video.

Final Project Standards

Evaluation: silhouette, Davies-Bouldin, Calinski-Harabasz, stability, explained variance, anomaly score analysis, and domain interpretation. **Required:** preprocessing pipeline, parameter justification, clear visualizations, GitHub README, Streamlit dashboard, PDF report, and reproducible results. **Important:** do not report ordinary classification accuracy as the main unsupervised-learning metric.

Amanullah Fazil S

Project Title	Customer Value Segmentation & Retention Analysis
Algorithms	K-Means Hierarchical Clustering
Dataset	Online Retail Dataset (UCI) Click to open the official dataset page

Objective

Transform raw retail transactions into customer-level behavioural features and discover meaningful customer groups. Compare K-Means with Hierarchical Clustering, profile each segment, and recommend retention actions for high-value, regular, new, and at-risk customers.

Tasks to Complete

- 1 Load the transaction data and inspect shape, data types, null values, duplicates, cancellations, negative quantities, and invalid prices.
- 2 Create TotalAmount and customer-level RFM features: Recency, Frequency, and Monetary Value. Add average order value and unique products purchased.
- 3 Visualize feature distributions and correlations. Apply suitable log transformation and StandardScaler or RobustScaler.
- 4 Train K-Means for $k = 2$ to 8. Use the elbow curve and silhouette score to select the best value of k .
- 5 Profile K-Means clusters using cluster size, median RFM values, spending level, and purchase frequency.
- 6 Apply Hierarchical Clustering. Compare ward, complete, and average linkage; create and interpret a dendrogram.
- 7 Compare both algorithms using Silhouette, Davies-Bouldin, Calinski-Harabasz, cluster balance, and business interpretability.
- 8 Visualize final clusters in two dimensions using PCA and assign meaningful business names to every segment.
- 9 Build a small Streamlit dashboard that accepts customer data, displays segments, and presents retention recommendations.

Udayanidhi S

Project Title	Wholesale Buyer Profile Discovery
Algorithms	K-Means Principal Component Analysis
Dataset	Wholesale Customers Dataset (UCI) Click to open the official dataset page

Objective

Discover purchasing patterns among wholesale buyers using annual spending across product categories. Use PCA to compress and explain the feature space, then use K-Means to form buyer profiles that can support inventory planning and targeted sales strategies.

Tasks to Complete

- 1 Load the dataset and inspect Channel, Region, Fresh, Milk, Grocery, Frozen, Detergents_Paper, and Delicassen features.
- 2 Perform EDA with summary statistics, skewness checks, box plots, pairwise correlations, and category spending distributions.
- 3 Treat Channel and Region as descriptive profile fields, not clustering targets. Log-transform skewed spending variables and scale them.
- 4 Apply PCA and report explained variance for each component. Determine how many components preserve at least 90% of the variance.
- 5 Interpret PCA loadings to explain which product categories drive the major components.
- 6 Train K-Means for $k = 2$ to 8 on both the scaled original features and PCA-transformed features.
- 7 Compare models using silhouette, Davies-Bouldin, Calinski-Harabasz, inertia, and cluster stability across random seeds.
- 8 Create 2D and 3D PCA cluster plots and describe each buyer group using its original spending profile.
- 9 Present business actions for every cluster and build an interactive dashboard for buyer-segment exploration.

Muthu Sankar V

Project Title	Dry Bean Morphology Pattern Discovery
Algorithms	Hierarchical Clustering DBSCAN
Dataset	Dry Bean Dataset (UCI) Click to open the official dataset page

Objective

Identify naturally occurring groups in bean morphology using shape and dimensional measurements. Compare Hierarchical Clustering with density-based DBSCAN, study irregular cluster structures, and investigate samples that appear as noise or unusual shapes.

Tasks to Complete

- 1 Load the dataset, inspect the 16 numerical morphology features, duplicates, missing values, and class distribution.
- 2 Remove the Class column during model training. Use it only after clustering as an external validation reference.
- 3 Visualize correlations, feature distributions, and possible outliers. Standardize all numerical features.
- 4 Apply Hierarchical Clustering on a representative sample. Compare ward, complete, average, and single linkage methods.
- 5 Create dendrograms and select a reasonable number of clusters using visual cuts and silhouette analysis.
- 6 Tune DBSCAN using a k-distance plot. Experiment with eps and min_samples and report the percentage of noise points.
- 7 Compare Hierarchical Clustering and DBSCAN using silhouette, Davies-Bouldin, cluster balance, and noise handling.
- 8 After training, use the known bean classes to compute ARI, NMI, and a cluster-vs-class cross-tab without retraining on labels.
- 9 Visualize clusters with PCA and explain which morphology features distinguish the discovered groups.

Lakshana R

Project Title	Bank Customer Campaign Segmentation
Algorithms	K-Means DBSCAN
Dataset	Bank Marketing Dataset (UCI) Click to open the official dataset page

Objective

Segment bank customers using demographic, financial, and campaign-interaction features without using the subscription outcome as a training label. Compare centroid-based and density-based clustering to identify stable customer groups and unusual campaign behaviour.

Tasks to Complete

- 1 Load the dataset, inspect categorical and numerical variables, identify unknown values, duplicates, and class imbalance in y .
- 2 Exclude y during clustering. Use it only later to understand how each discovered segment responds to campaigns.
- 3 Engineer useful features such as total previous contacts, contact recency groups, and campaign intensity.
- 4 Encode categorical variables with OneHotEncoder and scale numerical variables using a preprocessing pipeline.
- 5 Train K-Means for $k = 2$ to 10 and select k using elbow, silhouette, Davies-Bouldin, and cluster-size analysis.
- 6 Tune DBSCAN with a k -distance graph and multiple $\text{eps}/\text{min_samples}$ combinations. Measure cluster count and noise percentage.
- 7 Compare the two algorithms for quality, scalability, noise detection, and clarity of customer profiles.
- 8 Profile every cluster by age, balance, job, loan status, campaign contacts, and post-hoc subscription rate.
- 9 Create a dashboard with segment summaries and recommend a suitable communication strategy for each cluster.

Project Title	Human Activity Pattern Discovery from Smartphone Sensors
Algorithms	Principal Component Analysis K-Means
Dataset	Human Activity Recognition Using Smartphones (UCI) Click to open the official dataset page

Objective

Explore high-dimensional smartphone accelerometer and gyroscope data, reduce the feature space with PCA, and cluster similar motion patterns with K-Means. Examine whether unsupervised groups align with activities such as walking, sitting, standing, and laying.

Tasks to Complete

- 1 Load train and test sensor features, combine them carefully, and inspect the number of samples and high-dimensional variables.
- 2 Remove duplicate or near-constant features and keep activity labels completely outside the training feature matrix.
- 3 Standardize the sensor features and explain why scaling is essential before PCA and K-Means.
- 4 Apply PCA, create a scree plot, and select components that preserve approximately 90% to 95% of total variance.
- 5 Interpret the strongest loadings for the first components and explain the sensor patterns they represent.
- 6 Train K-Means for $k = 2$ to 10 in PCA space. Use elbow, silhouette, Davies-Bouldin, and Calinski-Harabasz scores.
- 7 Compare K-Means performance in the original scaled feature space and the reduced PCA feature space.
- 8 Use activity labels only after clustering to compute ARI, NMI, and a cluster-activity cross-tab.
- 9 Build a 2D or 3D interactive PCA visualization and summarize which activities are easy or difficult to separate.

Nitesh Varshan K K

Project Title	Network Intrusion Anomaly Discovery
Algorithms	Isolation Forest DBSCAN
Dataset	KDD Cup 1999 Network Data (UCI) Click to open the official dataset page

Objective

Detect unusual network connections without training a supervised attack classifier. Compare Isolation Forest and DBSCAN for anomaly discovery, analyze their false positives, and identify traffic characteristics that commonly produce high anomaly scores.

Tasks to Complete

- 1 Use the 10% subset or a reproducible sample. Inspect numerical, categorical, and connection-label columns.
- 2 Remove labels during training. One-hot encode protocol_type, service, and flag; scale numerical features appropriately.
- 3 Perform EDA on duration, bytes, connection counts, error rates, and feature skewness. Handle extreme values carefully.
- 4 Train Isolation Forest with contamination values such as 0.01, 0.03, 0.05, and 0.10. Compare anomaly-score distributions.
- 5 Apply DBSCAN after dimensionality reduction or feature selection. Tune eps with a k-distance curve and vary min_samples.
- 6 Compare anomaly counts, overlapping detections, execution time, memory usage, and sensitivity to parameter changes.
- 7 Use attack labels only after training to report precision, recall, F1, ROC-AUC or PR-AUC where appropriate.
- 8 Investigate at least 20 detected anomalies and identify the features that make them different from normal traffic.
- 9 Build a Streamlit anomaly dashboard with filters, score plots, traffic summaries, and a downloadable detection report.

Palani Bharathi

Project Title	Wine Chemical Profile Segmentation & Outlier Detection
Algorithms	K-Means Isolation Forest
Dataset	Wine Quality Dataset (UCI) Click to open the official dataset page

Objective

Group wines by physicochemical profile and detect samples with unusual chemical combinations. Use K-Means for interpretable wine-style segments and Isolation Forest for outlier discovery, then study how quality scores differ across the resulting groups.

Tasks to Complete

- 1 Load red and/or white wine data, inspect all chemical features, missing values, duplicates, and quality distribution.
- 2 Exclude quality during unsupervised training. Keep it only for post-hoc analysis of clusters and anomalies.
- 3 Perform EDA with histograms, box plots, correlation heatmap, and red-vs-white comparisons when both datasets are used.
- 4 Scale features and train K-Means for $k = 2$ to 8. Select k using elbow, silhouette, Davies-Bouldin, and stability.
- 5 Profile each cluster using acidity, residual sugar, sulphates, alcohol, density, and post-hoc quality distribution.
- 6 Train Isolation Forest with several contamination settings and analyze anomaly-score thresholds.
- 7 Inspect anomalous wines and explain which chemical measurements make them unusual.
- 8 Compare the purpose and output of clustering versus anomaly detection, including overlap between rare clusters and anomalies.
- 9 Create a visual dashboard and write recommendations for quality-control review without claiming a medical or safety decision.

Project Title	Tumor Feature Structure Discovery
Algorithms	Principal Component Analysis Hierarchical Clustering
Dataset	Breast Cancer Wisconsin Diagnostic (UCI) Click to open the official dataset page

Objective

Study the hidden structure of tumor measurement data without training a diagnostic classifier. Use PCA to explain dominant feature combinations and Hierarchical Clustering to discover groups, then compare those groups with diagnosis labels only as external validation.

Tasks to Complete

- 1 Load the dataset, remove ID-like columns, inspect the 30 numerical features, missing values, and diagnosis counts.
- 2 Keep diagnosis out of all unsupervised training steps. Standardize every numerical feature.
- 3 Perform PCA, plot explained variance, and choose the minimum components needed to preserve at least 90% variance.
- 4 Interpret component loadings and identify whether size, texture, smoothness, or concavity features dominate each component.
- 5 Apply Hierarchical Clustering in the selected PCA space and compare ward, complete, and average linkage.
- 6 Generate a dendrogram and choose the cluster cut using silhouette and dendrogram structure.
- 7 Evaluate cluster quality using silhouette, Davies-Bouldin, Calinski-Harabasz, and cluster balance.
- 8 After training, compare clusters with diagnosis using cross-tabs, ARI, NMI, and cluster-level proportions.
- 9 Create a careful scientific conclusion that describes pattern discovery only and does not present the system as clinical diagnosis.

Project Title	Household Energy Usage Pattern & Anomaly Detection
Algorithms	Hierarchical Clustering Isolation Forest
Dataset	Individual Household Electric Power Consumption (UCI) Click to open the official dataset page

Objective

Discover recurring daily electricity-consumption patterns and identify unusual high-load or irregular days. Aggregate minute-level measurements into meaningful daily features, compare hierarchical groups, and use Isolation Forest to flag abnormal consumption behaviour.

Tasks to Complete

- 1 Load the time-series data, parse Date and Time, replace missing markers, and inspect the measurement columns.
- 2 Resample minute data into hourly and daily summaries. Create peak load, average load, variance, night usage, and sub-meter shares.
- 3 Visualize daily, weekly, and monthly trends; inspect seasonality and correlations between total power and sub-metering.
- 4 Scale daily features and apply Hierarchical Clustering with ward, complete, and average linkage.
- 5 Create a dendrogram, select the number of daily-pattern clusters, and evaluate with silhouette and Davies-Bouldin scores.
- 6 Train Isolation Forest on daily features using multiple contamination settings and rank days by anomaly score.
- 7 Compare anomalous days with the discovered clusters and inspect the original hourly load curves for selected dates.
- 8 Describe each cluster as a usage pattern such as low-use, evening-peak, sustained-high, or weekend-style consumption.
- 9 Build an interactive dashboard showing load curves, clusters, anomalous dates, and practical energy-saving observations.